SIT452 – Cloud Computing and Virtualized Networking

Cloud Computing - Introduction

***Acknowledgment**: The presentations contains some figures and text from the following:*
Cloud Computing Theory and Practice book by Dan C. Marinescu
Information Storage and Management, John Wiely & Sons, Inc.
Cloud Networking Understanding Cloud-based Data Center Networks by Gary Lee

IP SAN

- A SAN that uses Internet Protocol (IP) for the transport of block-level data
- Uses IP-based protocols for communication, commonly:
    a. Internet SCSI (iSCSI)
    b. Fibre Channel over IP (FCIP)

IP SAN

- Existing IP-based network infrastructure can be leveraged
  - ✓ Reduced cost compared to deploying new FC SAN infrastructure
- IP network makes it possible to extend or connect SANs over long distances

- Many long-distance disaster recovery solutions already leverage IP-based network

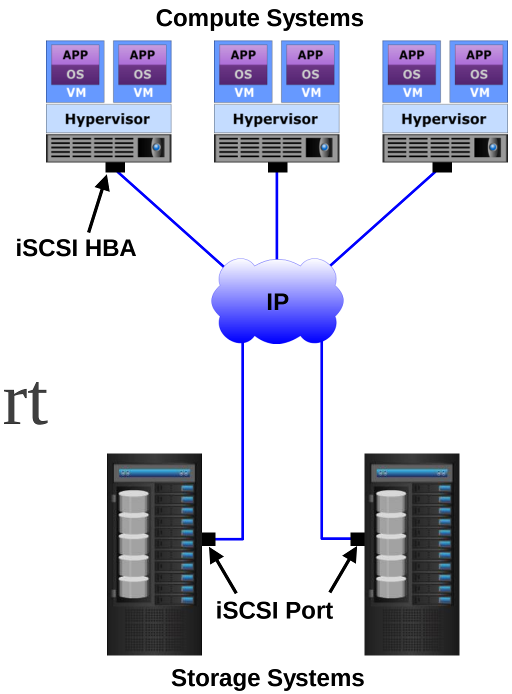
- Many robust and mature security options are available for IP network

iSCSI Overview

- IP-based protocol that enables transporting SCSI data over an IP network
- Encapsulates SCSI I/O into IP packets and transports them using TCP/IP

iSCSI Overview

- iSCSI initiators
  Example: iSCSI HBA
- iSCSI targets
  Example: Storage system with iSCSI port
- IP-based network
  Example: Gigabit Ethernet LAN



iSCSI Overview

Types of iSCSI Initiator

1. Standard NIC with software iSCSI adapter

- ✓ The software iSCSI adapter is an operating system (OS) or hypervisor kernel-resident software that uses an existing NIC of the compute system to emulate an iSCSI initiator.
- ✓ It is least expensive and easy to implement
- ✓ It requires only a software initiator for iSCSI functionality.
- ✓ Places additional overhead on the CPU.

iSCSI Overview

Types of iSCSI Initiator

2. TOE NIC performs TCP/IP processing

- ✓ Offloads the TCP/IP processing from the CPU of a compute system and leaves only the iSCSI functionality to the CPU.
- ✓ Improves performance, the iSCSI functionality is still handled by a software adapter that requires CPU cycles of the compute system.

iSCSI Overview

Types of iSCSI Initiator

3. iSCSI HBA

- Performs both iSCSI and TCP/IP processing
- Frees-up CPU cycles of compute system for business applications

iSCSI Overview

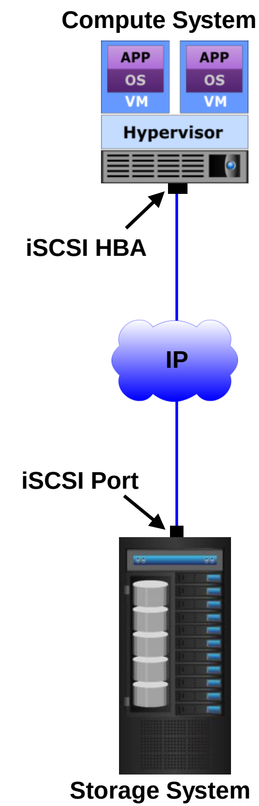
- The iSCSI implementations support two types of connectivity:

1. **Native iSCSI:**

✓ iSCSI initiators may be either directly attached to the iSCSI targets (iSCSI-capable storage systems) or connected through an IP-based network.

✓ FC components are not required for native iSCSI connectivity.

✓ After an iSCSI initiator is logged on to the network, it can access the available LUNs on the storage system.



iSCSI Overview

- The iSCSI implementations support two types of connectivity:

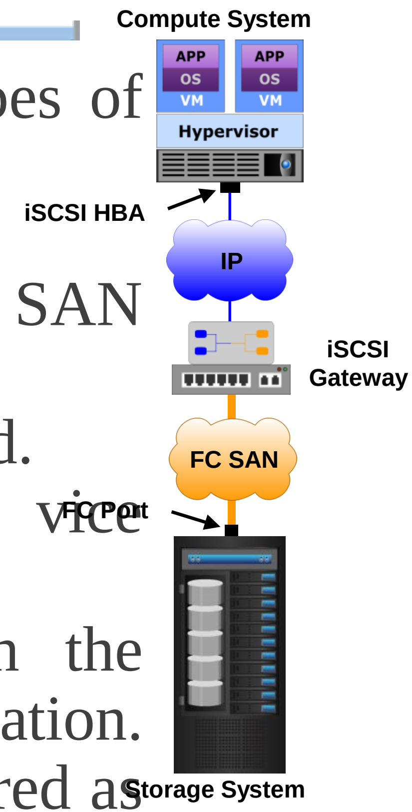
2. **Bridged iSCSI:**

✓ The storage systems remain in an FC SAN environment.

✓ A gateway or a multiprotocol router is used.

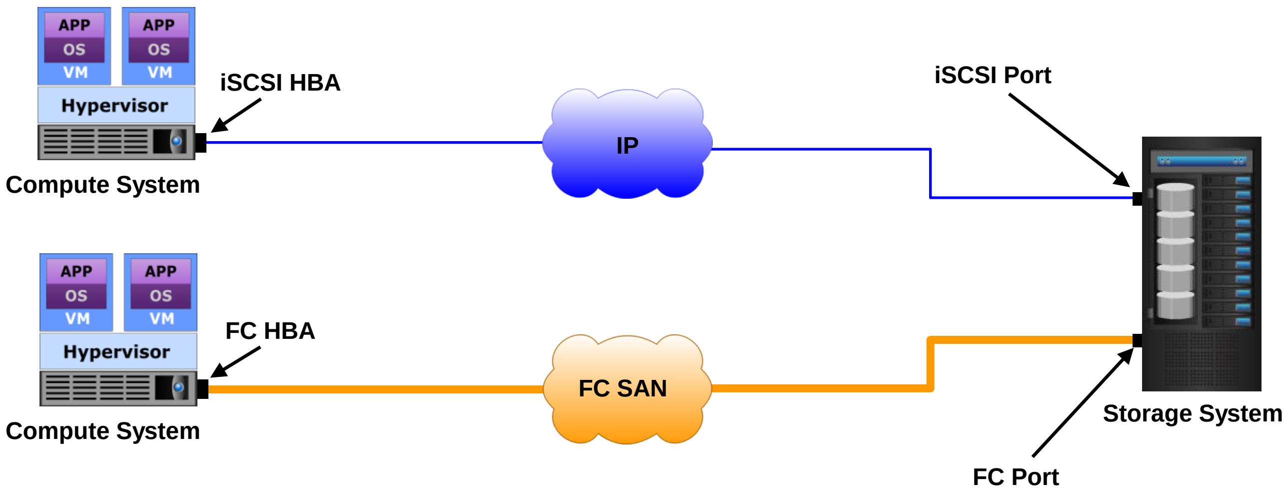
✓ Converts IP packets to FC frames and vice versa,

✓ The iSCSI initiator is configured with the gateway's IP address as its target destination. On the other side, the gateway is configured as an FC initiator to the storage system.

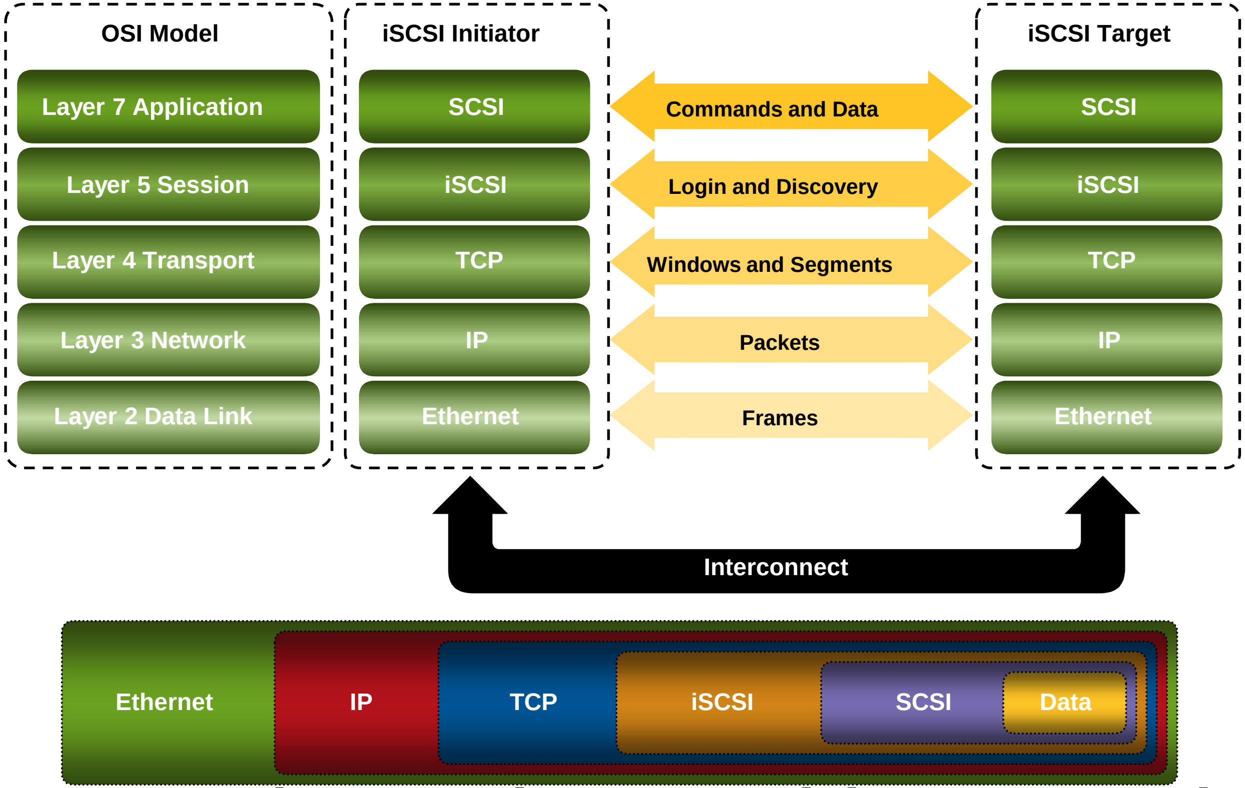


iSCSI Overview

- Storage system provides both FC and iSCSI ports
- Enable iSCSI and FC connectivity in the same environment
- No bridge device is needed



iSCSI Overview



- iSCSI is the session-layer protocol that initiates a reliable session between devices that recognize SCSI commands and TCP/IP. The iSCSI session-layer interface is responsible for handling login, authentication, target discovery, and session management.

iSCSI Overview

- iSCSI address is the path to iSCSI initiator/target, which is comprised of:
  - ✓ Location of iSCSI initiator/target
    - Combination of IP address and TCP port number

  - ✓ iSCSI name
    - Unique identifier for initiator/target in an iSCSI network

iSCSI Overview

- For iSCSI communication, initiator must discover location and name of targets on the network
- iSCSI discovery commonly takes place in two ways:

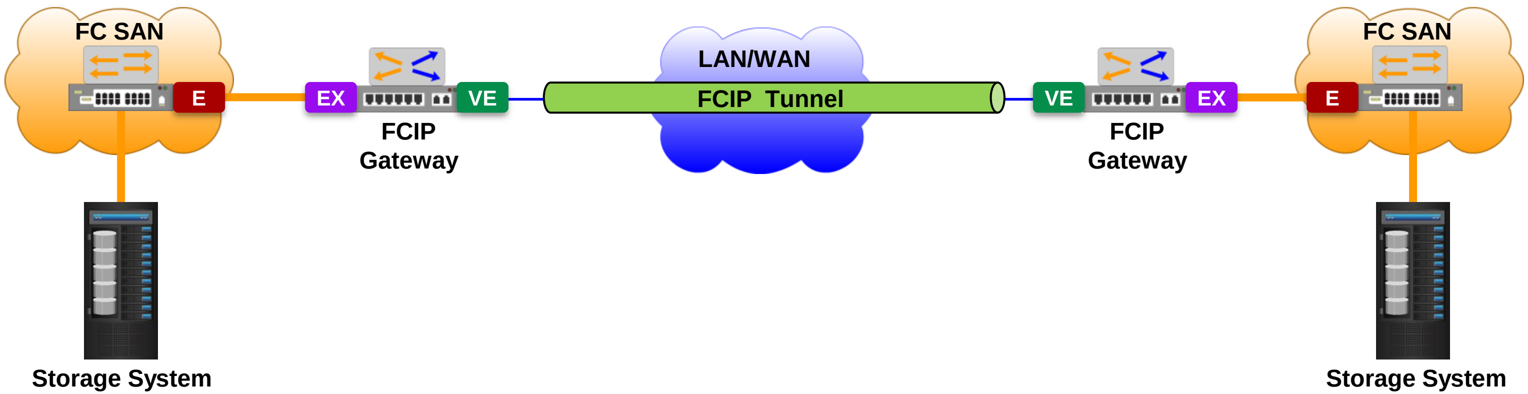
  - SendTargets discovery
    - ✓ Initiator is manually configured with the target’s network portal
    - ✓ Initiator issues SendTargets command; target responds with required parameters
  - Internet Storage Name Service (iSNS)
    - ✓ Initiators and targets register themselves with iSNS server
    - ✓ Initiator may query iSNS server for a list of available targets

iSCSI Overview

- iSNS discovery domains function in the same way as FC zones.
- Discovery domains provide functional groupings of devices (including iSCSI initiators and targets) in an IP SAN.
- The iSNS server is configured with discovery domains. For devices to communicate with one another, they must be configured in the same discovery domain.
- The iSNS server may send state change notifications (SCNs) to the registered devices. State change notifications (SCNs) inform the registered devices about network events that affect the operational state of devices such as the addition or removal of devices from a discovery domain.

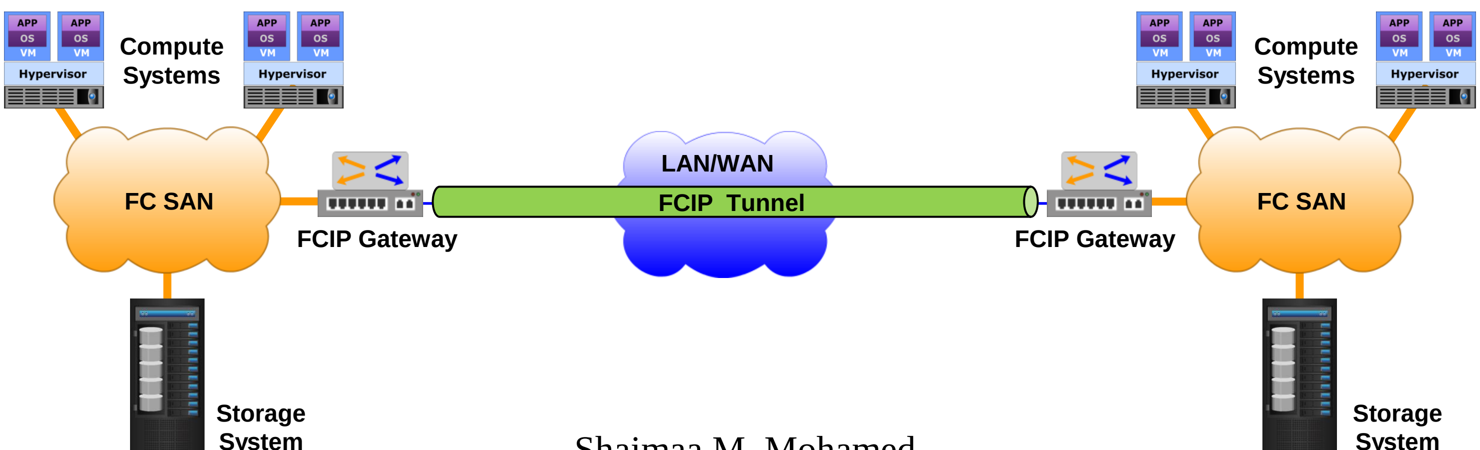
FCIP

- FCIP tunnel uses vendor-specific features to route network traffic between specific nodes without merging the fabrics
- EX_Port on FCIP gateway connects to E_Port of an FC switch
- Enables FC-FC routing without merging fabrics



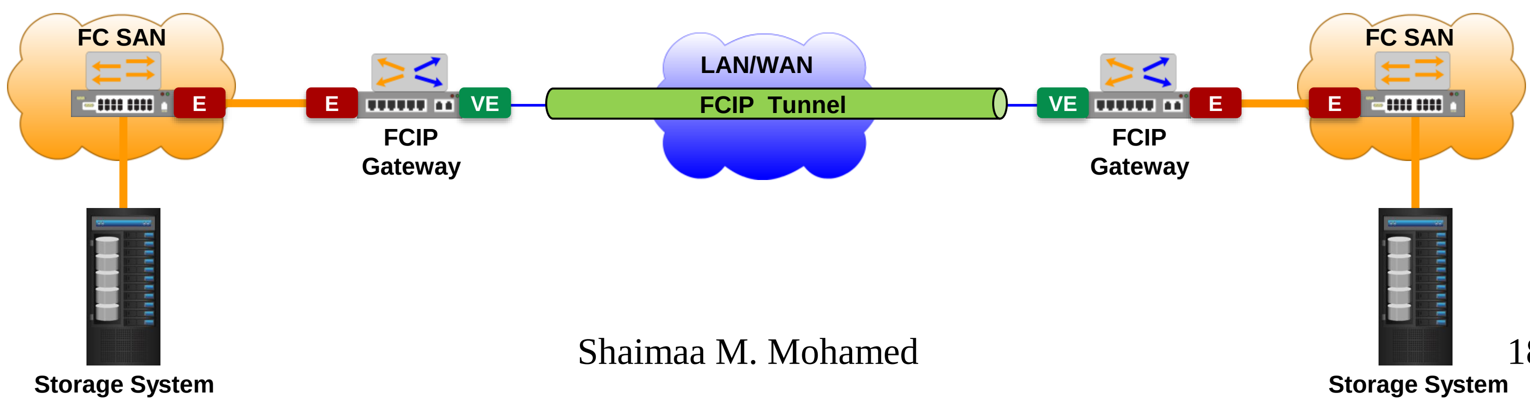
FCIP

- FCIP entity (e.g. FCIP gateway) is connected to each fabric to enable tunneling through an IP network
- An FCIP tunnel consists of one or more independent connections between two FCIP ports
  - Transports encapsulated FC frames over TCP/IP

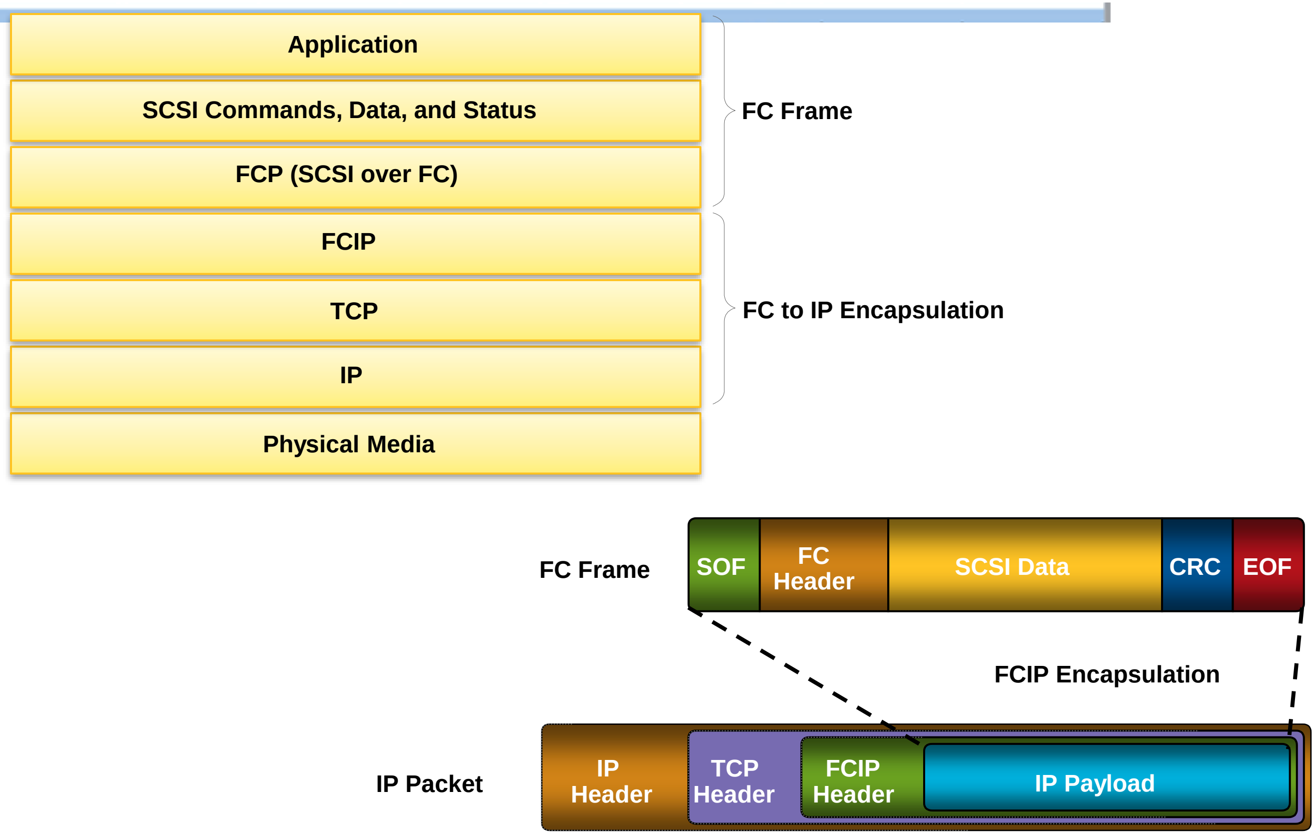


FCIP

- An FCIP tunnel may be configured to merge interconnected fabrics into a single large fabric.
- In the merged fabric, FCIP transports existing fabric services across the IP network.
- In this deployment, the E_Port on an FCIP gateway connects to the E_Port of an FC switch in the adjoined fabric.
- The FCIP gateway is also configured with a VE_port that behaves like an E_Ports, except that the VE_Port is used to transport data through an FCIP tunnel.
- The FCIP tunnel has VE_Ports on both ends. The VE_Ports establish virtual ISLs through the FCIP tunnel, which enable fabrics on either side of the tunnel to merge.



FCIP


Application
SCSI Commands, Data, and Status
FCP (SCSI over FC)
FCIP
TCP
IP
Physical Media
FC Frame
FC to IP Encapsulation
FC Frame
SOF
FC Header
SCSI Data
CRC
EOF
FCIP Encapsulation
IP Packet
IP Header
TCP Header
FCIP Header
IP Payload

Questions?